

Counting Numbers

We have already learnt numbers up to and across 99 in previous class which are as follows:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	

When we add 1 more in 99, we get 100 and read it as one hundred.

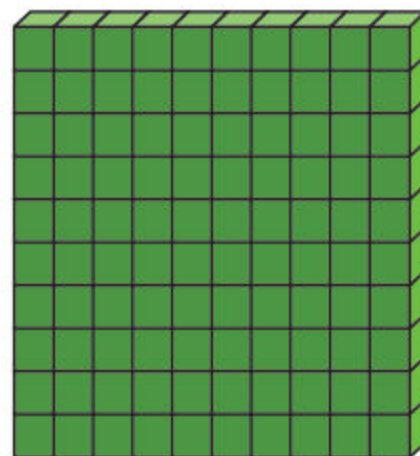
10 tens make 1 hundred →

i.e. 10 tens = 100

100 is a first 3-digit number.

In place value chart, we write it as:

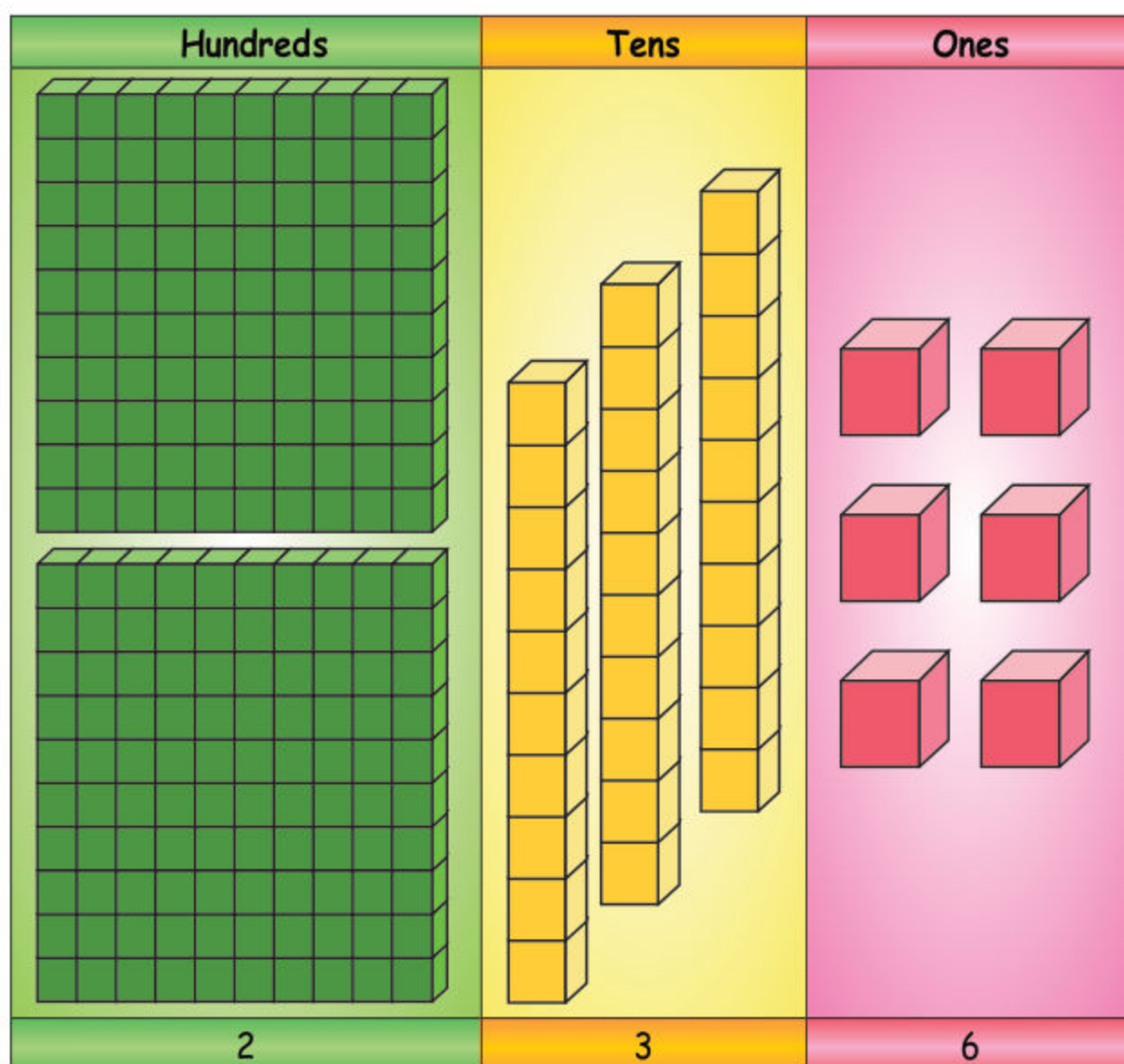
Hundreds	Tens	Ones
1	0	0



Unit 1: Numbers

Example 1

Count hundreds, tens and ones. Also write the number.



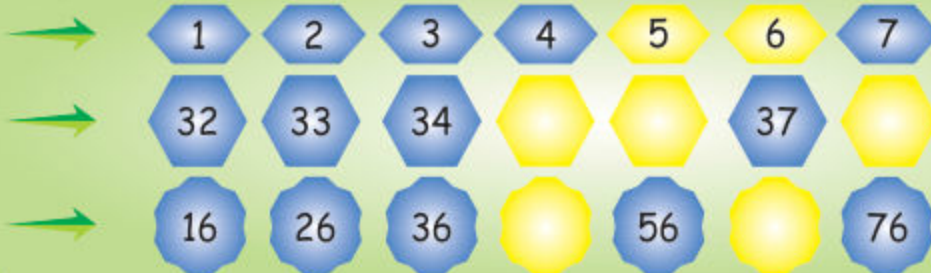
Number = 236

2 hundreds 3 tens 6 ones

Two hundred thirty six = 236

**Activity**

Count forward and write the missing numbers

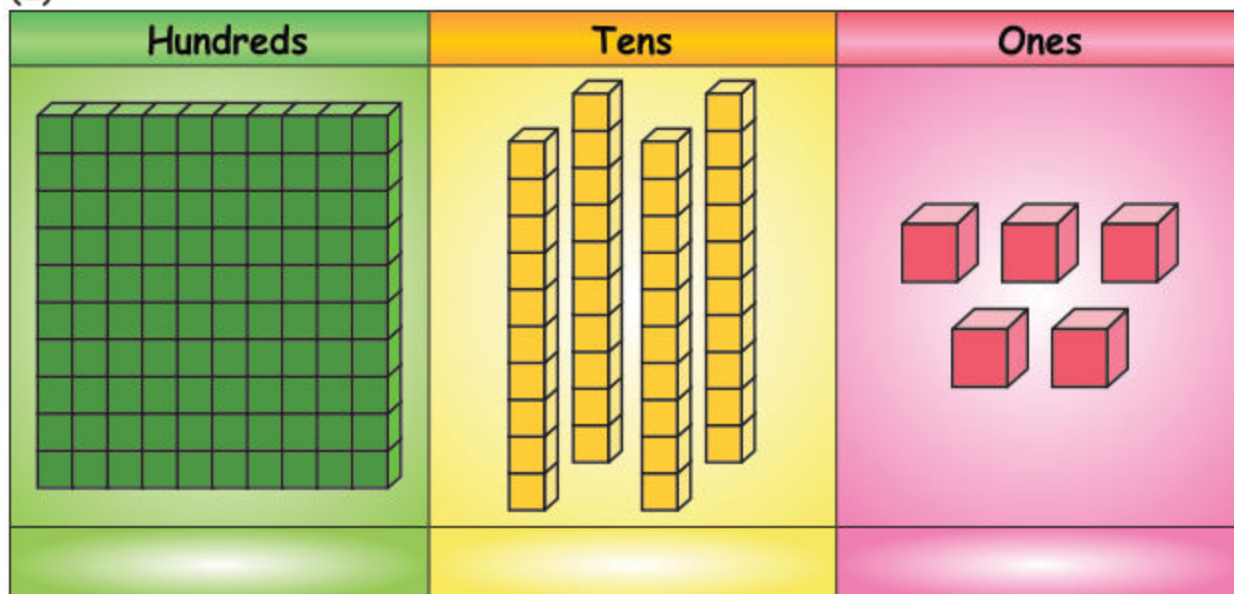
**Activity**

Count backward and write the missing numbers.

**Exercise 1**

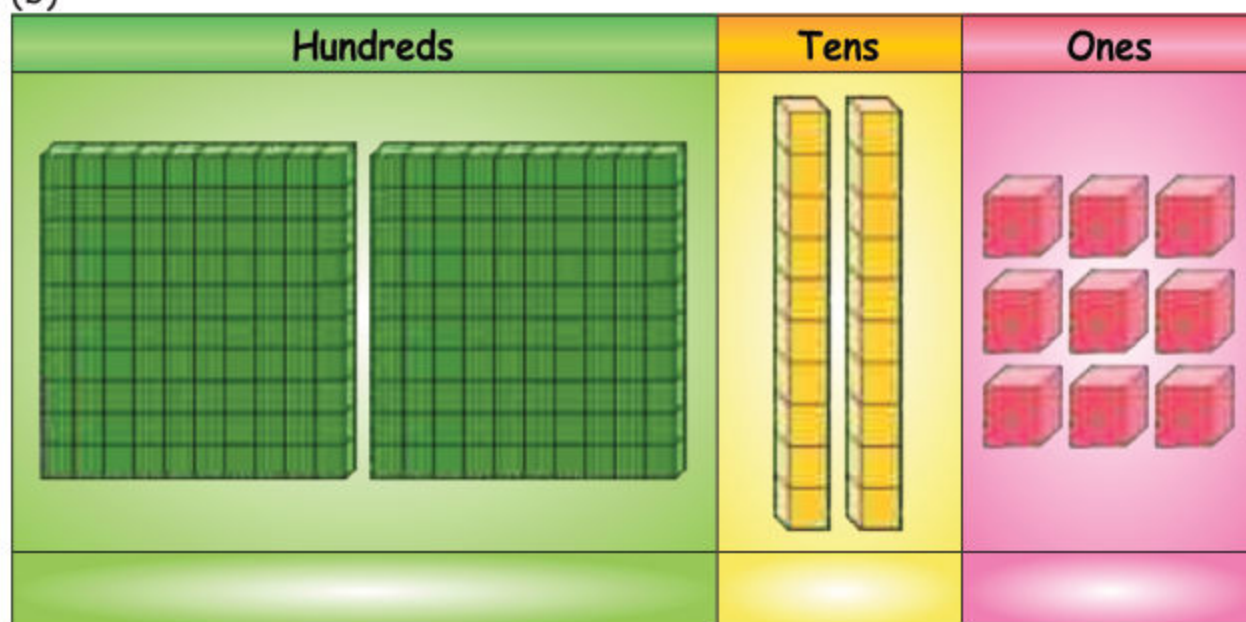
1. Count hundreds, tens and ones. Also write the numbers.

(a)



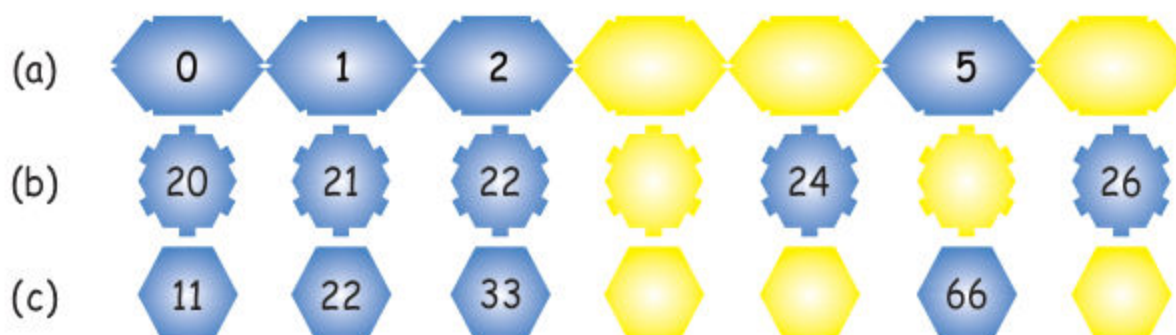
Number =

(b)

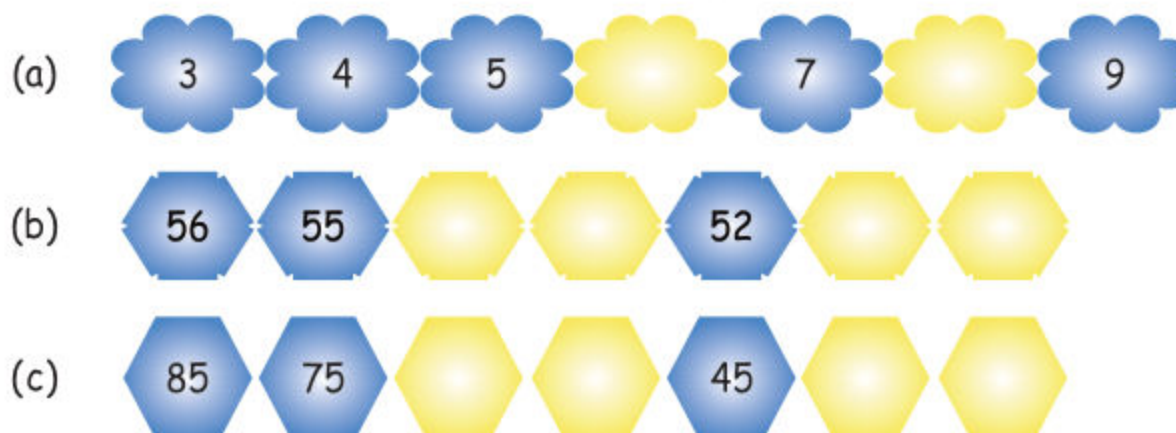


Number =

2. Count forward and write the missing numbers.



3. Count backward and write the missing numbers.



Reading and writing 3-digit numbers.

Read numbers from 100 to 299

100	101	102	103	104	105	106	107	108	109
110	111	112	113	114	115	116	117	118	119
120	121	122	123	124	125	126	127	128	129
130	131	132	133	134	135	136	137	138	139
140	141	142	143	144	145	146	147	148	149
150	151	152	153	154	155	156	157	158	159
160	161	162	163	164	165	166	167	168	169
170	171	172	173	174	175	176	177	178	179
180	181	182	183	184	185	186	187	188	189
190	191	192	193	194	195	196	197	198	199
200	201	202	203	204	205	206	207	208	209
210	211	212	213	214	215	216	217	218	219
220	221	222	223	224	225	226	227	228	229
230	231	232	233	234	235	236	237	238	239
240	241	242	243	244	245	246	247	248	249
250	251	252	253	254	255	256	257	258	259
260	261	262	263	264	265	266	267	268	269
270	271	272	273	274	275	276	277	278	279
280	281	282	283	284	285	286	287	288	289
290	291	292	293	294	295	296	297	298	299

Unit 1: Numbers

Reading and writing 3-digit numbers.

Read numbers from 300 to 499.

300	301	302	303	304	305	306	307	308	309
310	311	312	313	314	315	316	317	318	319
320	321	322	323	324	325	326	327	328	329
330	331	332	333	334	335	336	337	338	339
340	341	342	343	344	345	346	347	348	349
350	351	352	353	354	355	356	357	358	359
360	361	362	363	364	365	366	367	368	369
370	371	372	373	374	375	376	377	378	379
380	381	382	383	384	385	386	387	388	389
390	391	392	393	394	395	396	397	398	399
400	401	402	403	404	405	406	407	408	409
410	411	412	413	414	415	416	417	418	419
420	421	422	423	424	425	426	427	428	429
430	431	432	433	434	435	436	437	438	439
440	441	442	443	444	445	446	447	448	449
450	451	452	453	454	455	456	457	458	459
460	461	462	463	464	465	466	467	468	469
470	471	472	473	474	475	476	477	478	479
480	481	482	483	484	485	486	487	488	489
490	491	492	493	494	495	496	497	498	499

Reading and writing 3-digit numbers.

Read numbers from 800 to 999.

800	801	802	803	804	805	806	807	808	809
810	811	812	813	814	815	816	817	818	819
820	821	822	823	824	825	826	827	828	829
830	831	832	833	834	835	836	837	838	839
840	841	842	843	844	845	846	847	848	849
850	851	852	853	854	855	856	857	858	859
860	861	862	863	864	865	866	867	868	869
870	871	872	873	874	875	876	877	878	879
880	881	882	883	884	885	886	887	888	889
890	891	892	893	894	895	896	897	898	899
900	901	902	903	904	905	906	907	908	909
910	911	912	913	914	915	916	917	918	919
920	921	922	923	924	925	926	927	928	929
930	931	932	933	934	935	936	937	938	939
940	941	942	943	944	945	946	947	948	949
950	951	952	953	954	955	956	957	958	959
960	961	962	963	964	965	966	967	968	969
970	971	972	973	974	975	976	977	978	979
980	981	982	983	984	985	986	987	988	989
990	991	992	993	994	995	996	997	998	999



Activity Write the missing numbers

341	342								350
261	262								270
401	402								410
981									990
721									730



Activity Read and Trace the numbers from 1 to 50 in words.

In figures	In words	In figures	In words
1	One	11	Eleven
2	Two	12	Twelve
3	Three	13	Thirteen
4	Four	14	Fourteen
5	Five	15	Fifteen
6	Six	16	Sixteen
7	Seven	17	Seventeen
8	Eight	18	Eighteen
9	Nine	19	Nineteen
10	Ten	20	Twenty

In figures	In words	In figures	In words
21	Twenty one	36	Thirty six
22	Twenty two	37	Thirty seven
23	Twenty three	38	Thirty eight
24	Twenty four	39	Thirty nine
25	Twenty five	40	Forty
26	Twenty six	41	Forty one
27	Twenty seven	42	Forty two
28	Twenty eight	43	Forty three
29	Twenty nine	44	Forty four
30	Thirty	45	Forty five
31	Thirty one	46	Forty six
32	Thirty two	47	Forty seven
33	Thirty three	48	Forty eight
34	Thirty four	49	Forty nine
35	Thirty five	50	Fifty

Exercise 2

1. Write the missing numbers.

200	201			204			207			210
311	312				316			319		321
415			418				422			425
509				513				517		
625					630					634
895				899			902			
899			902				906			909
989		991				995				

2. Match the numbers in words with numbers in figures.

In words	In figures
Nineteen	37
Twelve	49
Twenty three	12
Twenty	27
Thirty seven	50
Forty nine	19
Twenty seven	20
Fifty	23

3. Write the numbers in figures.

In words	In figures
Eighteen	18
Twelve	
Twenty two	
Thirty eight	
Forty three	
Forty	
Thirty nine	
Seven	

4. Write the numbers in words.

In figures	In words
47	Forty seven
36	
11	
17	
26	
44	
8	
30	

Recognizing the place value of a number

We know that 100 is a first 3-digit number. Its place value chart is

Hundreds	Tens	Ones
1	0	0

Similarly, each 3-digit number has three place values

For example, the place values of 537 are shown below



The place value of 7 is 7 ones = 7

The place value of 3 is 3 tens = 30

The place value of 5 is 5 hundreds = 500

So, $537 = 500 + 30 + 7$

Example:

Identify the place value of encircled digit.

5 (6) 7

The place value of 6 is 6 tens = 60

(9) 7 8

The place value of 9 is 9 hundreds = 900

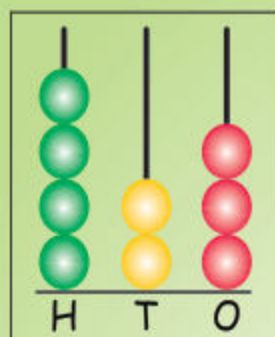
4 2 (3)

The place value of 3 is 3 ones = 3



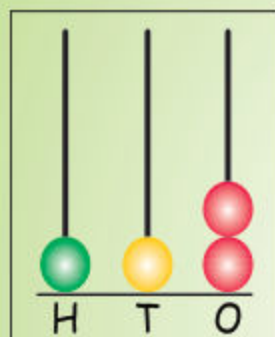
Activity

Count hundreds, tens, ones and write the numbers.



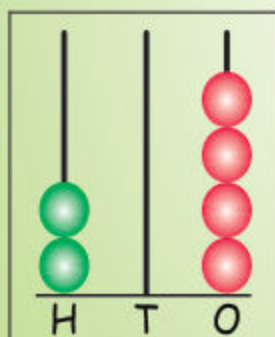
Hundreds	Tens	Ones
4	2	3

Number = 423



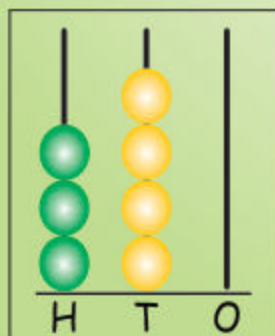
Hundreds	Tens	Ones

Number =



Hundreds	Tens	Ones

Number =



Hundreds	Tens	Ones

Number =

Exercise 3

1. Write the place value of encircled digit.

- | | | | | | | | |
|--|---|---|---|--|---|---|---|
| (a) <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>3</td><td>5</td><td>9</td></tr></table> 5 tens = 50 | 3 | 5 | 9 | (d) <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>6</td><td>9</td><td>8</td></tr></table> <table border="1" style="display: inline-table; vertical-align: middle; width: 100px; height: 20px;"></table> | 6 | 9 | 8 |
| 3 | 5 | 9 | | | | | |
| 6 | 9 | 8 | | | | | |
| (b) <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>7</td><td>1</td><td>3</td></tr></table> <table border="1" style="display: inline-table; vertical-align: middle; width: 100px; height: 20px;"></table> | 7 | 1 | 3 | (e) <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>2</td><td>4</td><td>6</td></tr></table> <table border="1" style="display: inline-table; vertical-align: middle; width: 100px; height: 20px;"></table> | 2 | 4 | 6 |
| 7 | 1 | 3 | | | | | |
| 2 | 4 | 6 | | | | | |
| (c) <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>5</td><td>9</td><td>8</td></tr></table> <table border="1" style="display: inline-table; vertical-align: middle; width: 100px; height: 20px;"></table> | 5 | 9 | 8 | (f) <table border="1" style="display: inline-table; vertical-align: middle;"><tr><td>8</td><td>0</td><td>1</td></tr></table> <table border="1" style="display: inline-table; vertical-align: middle; width: 100px; height: 20px;"></table> | 8 | 0 | 1 |
| 5 | 9 | 8 | | | | | |
| 8 | 0 | 1 | | | | | |

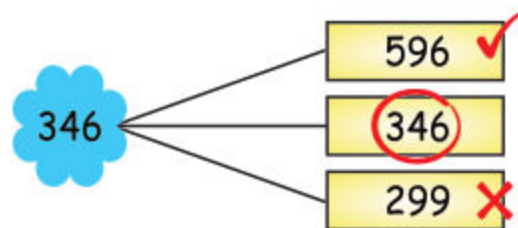
2. Count hundreds, tens, ones and write the numbers.

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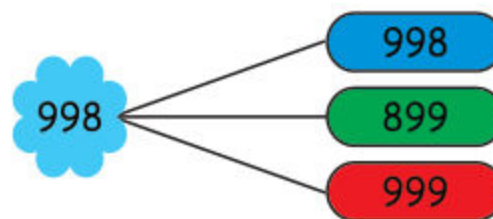
Comparing and ordering number

Example 1:

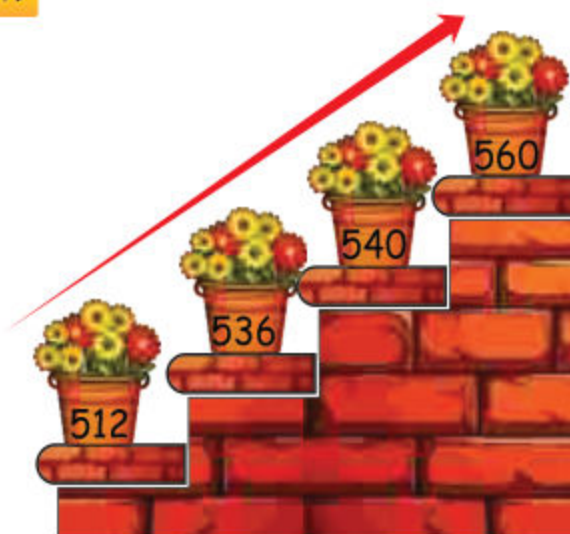
Tick the number which is more, cross the number which is less and encircle the number which is same as to given number.

**Example 2:**

Colour red the greater, colour green the smaller and colour blue the number which is equal to given number.

**Ascending and Descending order**

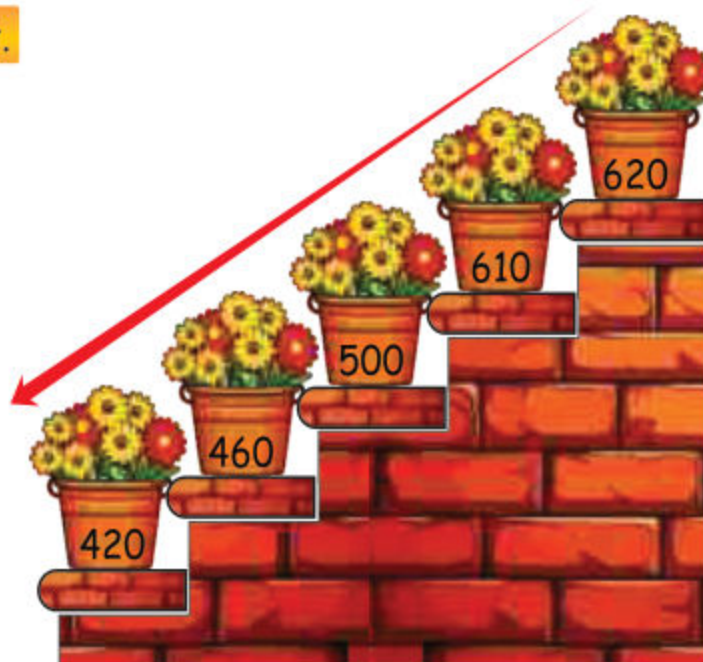
Ascending order means **increasing order** or arranging numbers from **smallest to largest**.



Ascending order: 512, 536, 540, 560

Unit 1: Numbers

Descending order means decreasing order or arranging numbers from largest to smallest.



Descending order: 620, 610, 500, 460, 420

Example 1:

Write the given numbers in ascending order

526 , 176 , 985 , 645 , 396

Ascending order: 176, 396, 526, 645, 985

Example 2:

Write the given numbers in descending order

127 , 476 , 896 , 645 , 255

Descending order: 896, 645, 476, 255, 127

Exercise 4

1. Tick the number which is more, cross the less number and encircle the number which is same as to given number.

(a)	236	512 ✓	236	117 ✗
(b)	525	525	496	552
(c)	815	684	920	815
(d)	305	503	305	130
(e)	114	108	141	114

2. Colour red for greater, colour green for smaller and colour blue for equal to given number.

(a)	419	914	419	276
(b)	675	675	705	576
(c)	829	829	452	928
(d)	800	825	800	450
(e)	101	100	102	101

3. Write the numbers in ascending order.



152,200,432,565



4. Write the numbers in descending order.



601,596,316,205



Rounding numbers

Rounding a number to the nearest tens

Rounding up

5 ↑

When the digit at ones place is 5 or more

We increase one more ten and place 0 at the position of ones.

Rounding down

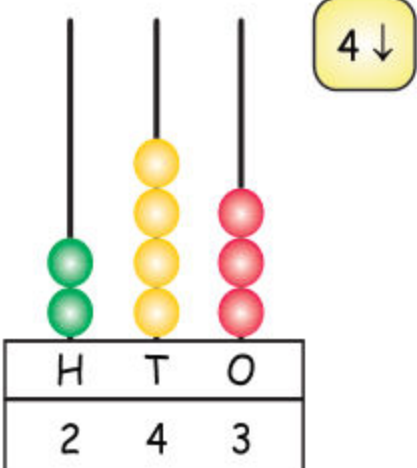
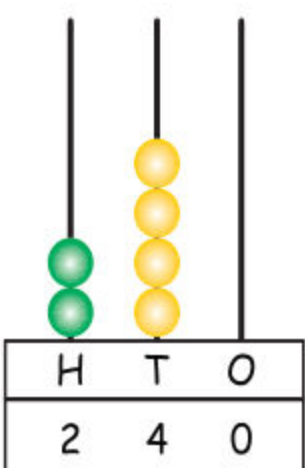
4 ↓

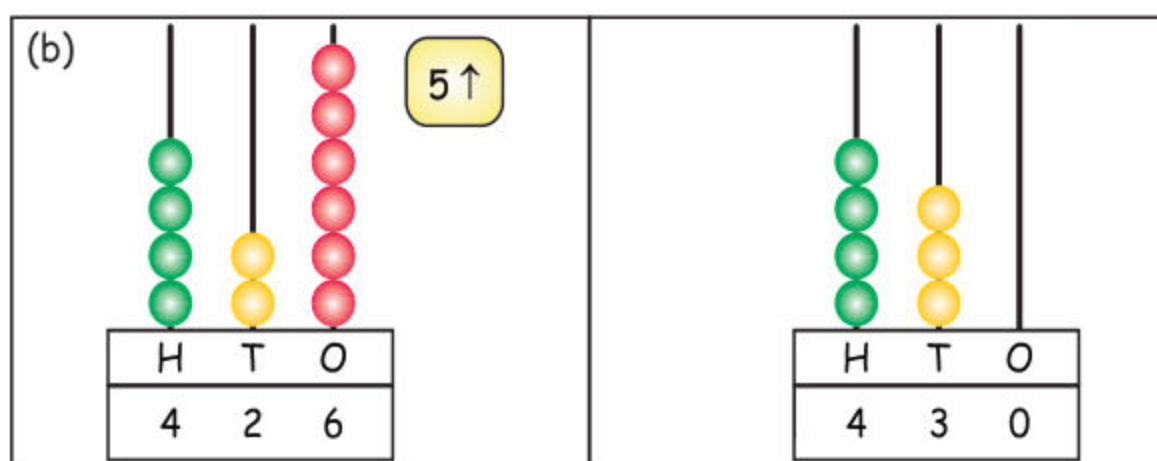
When the digit at ones place is 4 or less

We do not increase tens and write 0 at the position of ones.

Example 1:

Round the given numbers to the nearest tens.

Given numbers	Rounded to the nearest tens												
(a)  <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>2</td><td>4</td><td>3</td></tr></table>	H	T	O	2	4	3	 <table><tr><td>H</td><td>T</td><td>O</td></tr><tr><td>2</td><td>4</td><td>0</td></tr></table>	H	T	O	2	4	0
H	T	O											
2	4	3											
H	T	O											
2	4	0											

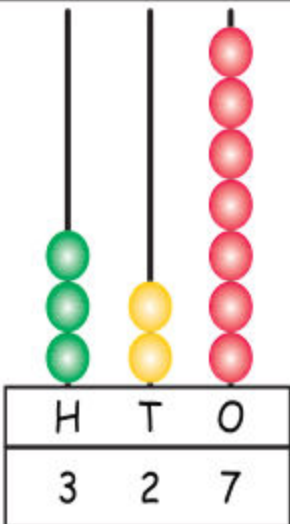
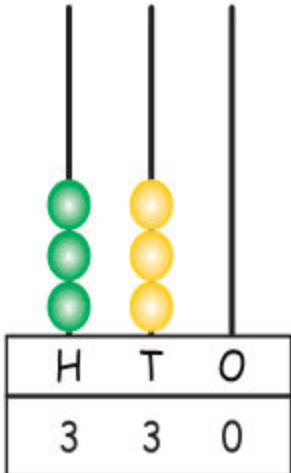
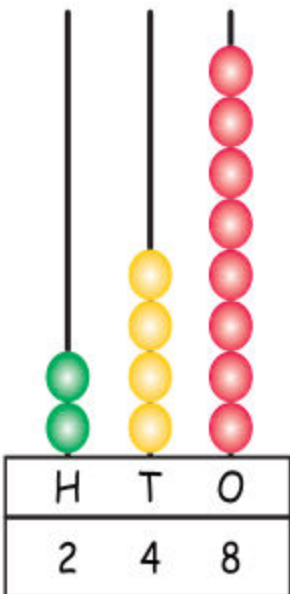
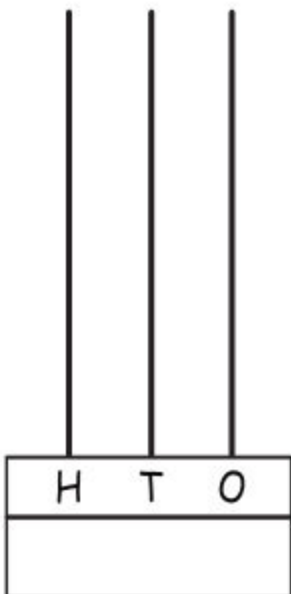

Example 2:

Round the given numbers to the nearest tens.

Given number			Rounded to the nearest tens		
(a)					
Hundreds	Tens	Ones	Hundreds	Tens	Ones
1	2	6	1	3	0
(b)					
Hundreds	Tens	Ones	Hundreds	Tens	Ones
1	2	4	1	2	0

Exercise 5

1. Round the given numbers to the nearest tens.

Given number	Rounded to the nearest tens
(a)  5 ↑	
(b) 	

(c)

H	T	O
2	3	2

H	T	O

2. Round the given numbers to the nearest tens.

Given number		Rounded to the nearest tens
(a) 5 6 9	5 ↑	5 7 0
(b) 3 9 2		
(c) 1 0 5		
(d) 2 3 7		
(e) 6 2 1		

Ordinal numbers

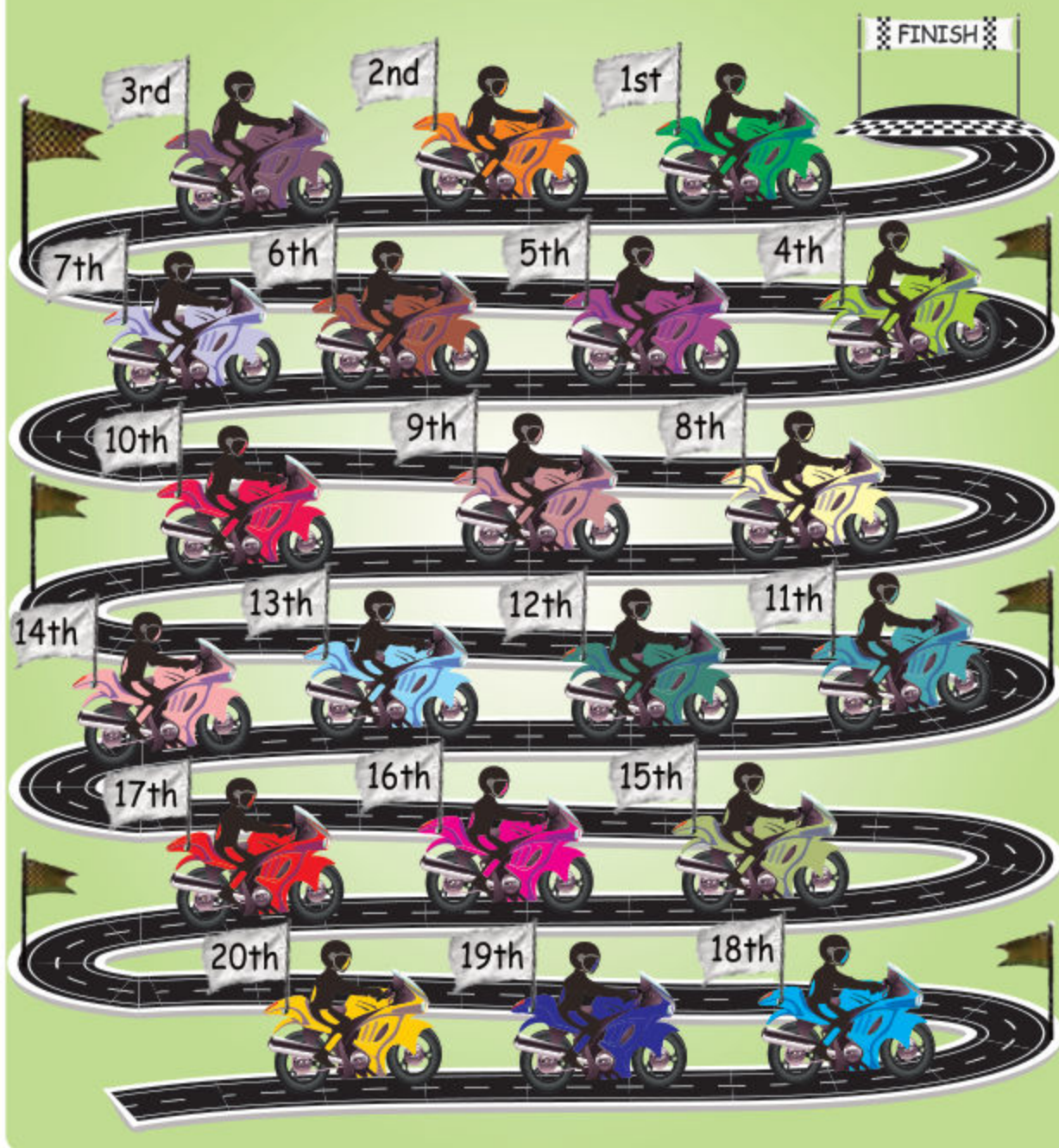
In previous class, we learnt to recognize the position of objects, using ordinal numbers up to 10.

Now, we learn ordinal numbers up to 20.



Activity

Read the positions of the bikes.





Activity Read the ordinal numbers up to 20.

Ordinal numbers	In words	Ordinal numbers	In words
1st	First	11th	Eleventh
2nd	Second	12th	Twelfth
3rd	Third	13th	Thirteenth
4th	Fourth	14th	Fourteenth
5th	Fifth	15th	Fifteenth
6th	Sixth	16th	Sixteenth
7th	Seventh	17th	Seventeenth
8th	Eighth	18th	Eighteenth
9th	Ninth	19th	Nineteenth
10th	Tenth	20th	Twentieth

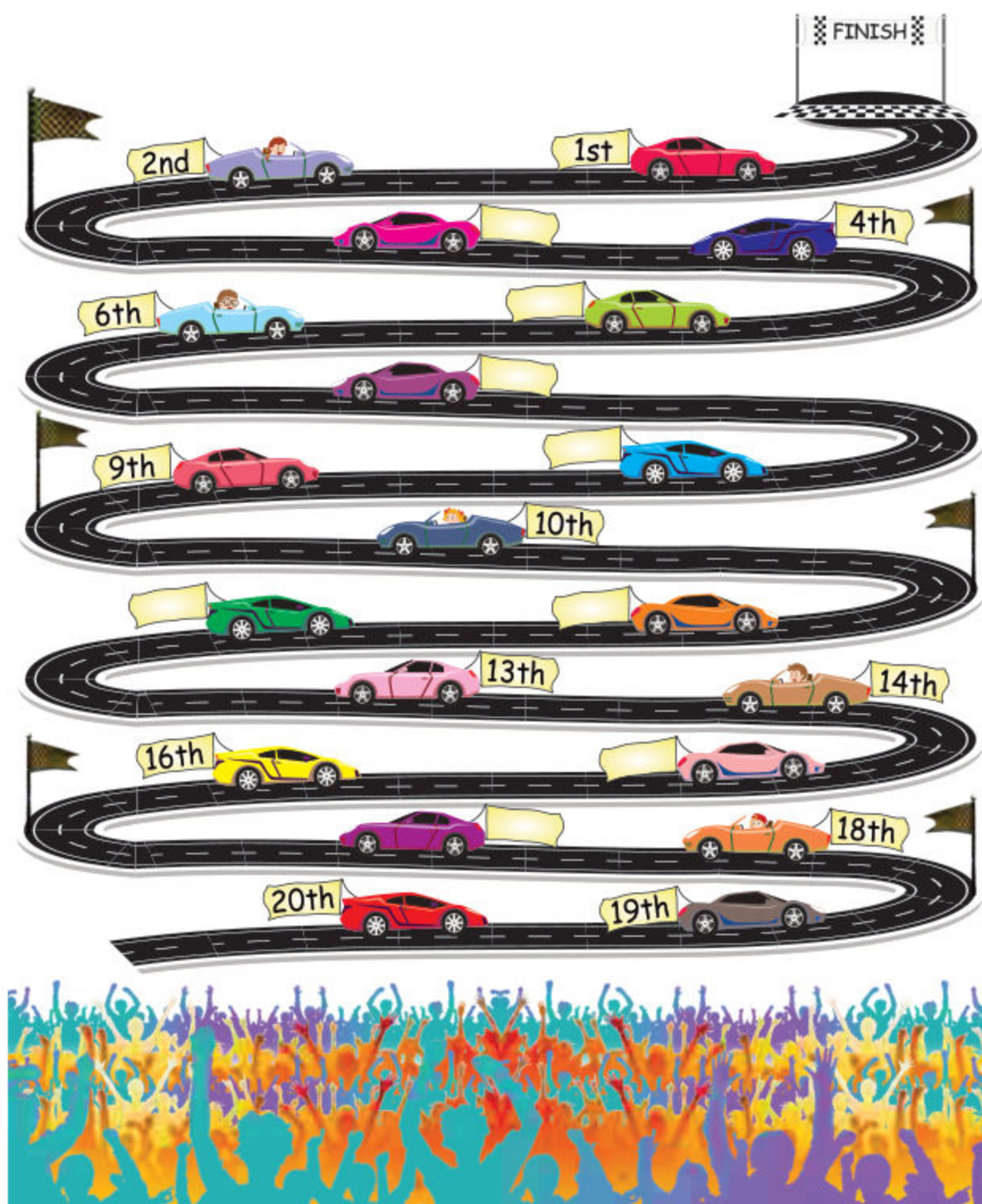


Activity Match the following.

Ordinal numbers	In words	Ordinal numbers	In words
1st	Fifth	11th	Fourteenth
2nd	Tenth	12th	Sixteenth
3rd	Sixth	13th	Seventeenth
4th	Seventh	14th	Nineteenth
5th	Eighth	15th	Eighteenth
6th	Ninth	16th	Eleventh
7th	Second	17th	Twelfth
8th	First	18th	Twentieth
9th	Third	19th	Fifteenth
10th	Fourth	20th	Thirteenth

Exercise 6

1. Look at the positions of the vehicles and write the missing position.



2. Write down the ordinal numbers for English letters (left to right)

A	B	C	D	E
1st	2nd			5th
F	G	H	I	J
6th			9th	
K	L	M	N	O
11th				
P	Q	R	S	T
	17th			

Roman numbers

We have learnt numbers up to 999. Besides these numbers we also use Roman numbers in different ways. They were mostly used in ancient Rome.

Nowadays roman numbers are usually used in watches, wall clocks, books and examination papers etc.

Roman numbers from 1 to 12 are based on three symbols or numerals which are as under:



Numeral in small letters	Numeral in capital letters	Value
i	I	1
v	V	5
x	X	10

Read Roman numbers from 1 to 12 which are as follows:

Number	Roman Number	
	Capital	Small
1	I	i
2	II	ii
3	III	iii
4	IV	iv
5	V	v
6	VI	vi
7	VII	vii
8	VIII	viii
9	IX	ix
10	X	x
11	XI	xi
12	XII	xii

Note: There is no Roman number for zero.

Exercise 7

1. Write in Roman numbers.

Number	Roman Number
3	III
8	
12	
1	
5	
4	

2. Write in Arabic numbers.

Roman Number	Number
II	2
VI	
VII	
IX	
XI	

3. Write the missing Roman numbers

I	II				VI
VII			X		

4. Match the following.

Number	Roman Number
5	iii
9	ii
12	viii
3	v
11	ix
6	xii
2	vi
8	xi

Unit

2

NUMBER OPERATIONS

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ find, recall and use addition and subtraction facts to 100.
- ✓ add and subtract numbers mentally and in written form (with and without regrouping) including: 3-digit number and a 1-digit number. 3-digit number and tens 3-digit number and a 2-digit number. Two 3-digit numbers
- ✓ estimate the answer to an addition and subtraction question. (using various approaches)
- ✓ recognize even and odd numbers
- ✓ count and write in 3s, 4s, 5
- ✓ recognize counting in 3s, 4s as multiplication tables of three and four.
- ✓ recognize multiplication as repeated addition and develop multiplication tables (times tables) for 2, 3, 4, 5 and 10.
- ✓ write multiplication statements (i.e., sentences) using concrete and pictorial representations.
- ✓ recognize using concrete and pictorial representations that the multiplication of any two numbers can be done.
- ✓ multiply mentally and in written form using the multiplication tables that they know: 1-digit number by another 1-digit number.
- ✓ multiply mentally and in written form using the multiplication tables that they know: 2-digit number by a 1-digit number using a multiplication grid.
- ✓ multiply a number with 0 and 1.
- ✓ recognize division as repeated subtraction through concrete and pictorial representation
- ✓ write division statements (i.e., sentences) using concrete and pictorial representations
- ✓ recognize using concrete and pictorial representation that the division of one number by another cannot be done in any order
- ✓ divide mentally and in written form: 1-digit number by another 1-digit number (without remainder) 2-digit number by a 1-digit number (without remainder)
- ✓ solve real-life word problems involving multiplication and division using any method (for instance materials, repeated addition/subtraction, groups, arrays, mental and or written methods).
- ✓ solve real-life word problems (including Pakistani currency) involving addition, subtraction, multiplication and division.